
3AL Tutorial 10 2024

1. Let G be a group with order $p^n m$ where $n \geq 1$, p is a prime with $p \nmid m$. Suppose H is a Sylow p -subgroup of G .
 - (a) Prove that H is a Sylow p -subgroup of $N_G(H)$ and that it is the only Sylow p -subgroup of $N_G(H)$.
 - (b) Let $a \in G$ such that $|a| = p^k$ where $k \geq 1$. Prove that if $aHa^{-1} = H$, then $a \in H$.
2. Let G be a finite group and H a normal p -subgroup. Show that if K is a Sylow p -subgroup of G , then H is a subgroup of K .
3. Prove the following:

Frattini's Argument. *Let G be a finite group with normal subgroup H . If K is a Sylow p -subgroup of H , then $G = HN_G(K)$.*

HINT: Since H is normal in G , we have $gKg^{-1} \subseteq H$ for all $g \in G$. Explain why this is true.

4. Let G be a finite group and p be a prime that divides $|G|$. Prove or disprove: If H and K are Sylow p -subgroups then H and K are isomorphic.
5. Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Suppose that H is a normal subgroup of G such that $|G : H|$ is coprime to p . Show that H contains all Sylow p -subgroups of G .
6. Let G be a group of order $90 = 2 \times 3^2 \times 5$. Suppose P is a Sylow 5-subgroup of G that is not normal. Determine the number of conjugates of P .
7. Prove that a group of order 200 has a nontrivial proper normal subgroup.
8. Use the Third Sylow Theorem to show that a group of order 18 has a nontrivial proper normal subgroup.
9. Let G be a group of order $56 = 2^3 \times 7$.
 - (a) Determine the possibilities for n_2 and n_7 .
 - (b) Determine the number of elements of order 7 if $n_7 \neq 1$.

(c) Show that either $n_2 = 1$ or $n_7 = 1$.

Extra problems

(I) Let P be a Sylow p -subgroup of a group G and H be a subgroup of G such that $N_G(P) \subseteq H$. Show that $N_G(H) = H$.

HINT: Show that, for $a \in N_G(H)$, we have $aPa^{-1} \subseteq H$. Then use the Second Sylow Theorem.